



Name: _____
Class: _____
Date: _____

Mr. Rawson • GCSE Physics

Charge, Current and Resistance

Key concept

You have built circuits, measured them, and on Friday you found how resistance depends on the length of a wire. What you have not done yet is the **language** — the symbols everyone draws circuits with, and the two equations that tie current, charge, potential difference and resistance together.

That is today. Three ideas, and you have already met all three in the laboratory without naming them.

1 Circuit symbols

A circuit diagram is not a picture of the apparatus — it is a **standard code**. Every physicist in the world draws a cell the same way, which is why a diagram drawn in this room can be built by someone who has never spoken to you.

Fill in what each one is. The blue sheet has all of them if you get stuck.

Symbol	Name	Symbol	Name
	<i>cell</i>		<i>battery</i>
	<i>switch (open)</i>		<i>lamp / bulb</i>
	<i>resistor</i>		<i>variable resistor</i>
	<i>ammeter</i>		<i>voltmeter</i>
	<i>diode</i>		<i>LED</i>
	<i>fuse</i>		<i>thermistor</i>

Now the other way round. Draw each symbol from memory — no looking.



cell



lamp



ammeter



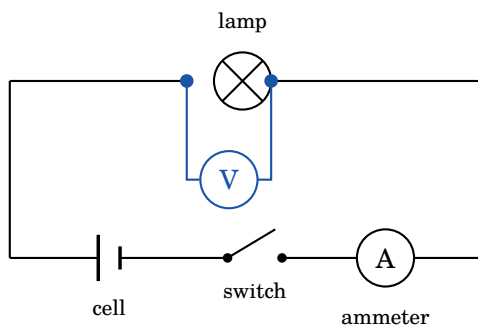
resistor

Two that students confuse, every year.

- **Cell and battery.** A *battery* is two or more cells joined together. One long line and one short line is a cell; the pattern repeated is a battery.
- **Ammeter and voltmeter** are both a circle with a letter, so the symbol will not help you.

Where they go is what tells them apart — and you used both on Friday.

2 Reading a circuit diagram



1) **State** whether this circuit is series or parallel, and how you can tell.

Series. There is only one loop, so there is only one path the current can take — no branches.

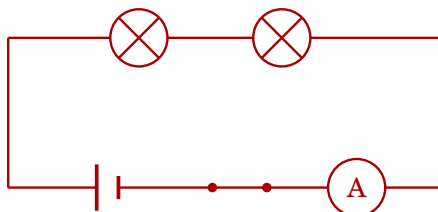
2) **Explain** why the ammeter is drawn *in* the loop but the voltmeter is drawn *across* the lamp.

An ammeter measures the current through the component, so all of the current must pass through it — it goes in series, in the loop. A voltmeter measures the difference in potential between two points, so it must be connected to both sides of the component — in parallel, across it.

3) **Predict** what the ammeter reads if the switch is left open.

Zero. An open switch breaks the loop, so the circuit is not complete and no charge can flow.

Now draw one. A cell, a closed switch, two lamps in series, and an ammeter reading the current through them. Pencil — you will want to change it.



One loop; both lamps and the ammeter in it. The ammeter may sit anywhere in the loop — the current is the same everywhere, so every position is correct. A closed switch is drawn as a straight connection between the two dots.

3 Charge and current

Key concept

Charge does not flow on its own. For charge to flow around a closed circuit, the circuit must contain a **source of potential difference** — a cell, a battery, or a power supply. That is what pushes it.

Electric current is the flow of electrical charge. More precisely, the size of the current is the **rate** of flow of charge — how much charge passes a point each second.

$$Q = It \quad \text{charge flow (C) = current (A) } \times \text{ time (s)}$$

Q	charge flow	measured in coulombs , C
I	current	measured in amperes , A (“amps” is accepted)
t	time	measured in seconds , s

4) **State** what is meant by *electric current*.

The rate of flow of electrical charge.

5) **Explain** why a circuit with no cell in it carries no current, even if every connection is perfect.

There is no source of potential difference, so there is nothing to push the charge around the circuit. A complete loop on its own is not enough.

Worked example

A current of 0.40 A flows through a lamp for 25 seconds. Calculate the charge that flows.

$$Q = It = 0.40 \times 25 = \mathbf{10\ C}$$

t is in SECONDS. Always. A question that says “for 2 minutes” is not giving you $t = 2$. Convert first, and write the conversion down where a marker can see it — that is where the method mark lives.

$$\mathbf{1\ minute = 60\ s} \quad \cdot \quad \mathbf{5\ minutes = 300\ s} \quad \cdot \quad \mathbf{1\ hour = 3600\ s}$$

Current in a single loop

Key concept

A current has the same value at every point in a single closed loop.

You measured this yourselves. In the series circuit you built, the ammeter read the same wherever you put it — before the first lamp, between the lamps, after the second.

6) **Explain** why the current must be the same at every point in a single closed loop.

Charge is not used up as it goes round the circuit, and it cannot leave the wire or pile up anywhere. So whatever amount of charge passes one point each second must pass every other point each second too — which means the current is the same everywhere.

Careful. It is the *current* that is shared out in a parallel circuit — but each separate branch is still its own path. The rule above is about a **single** loop.

4 Potential difference and resistance

Key concept

The current through a component depends on two things: the potential difference across it, and the resistance of the component.

The greater the resistance, the smaller the current — for a given potential difference. That is the sentence, and it is worth learning in those words.

$$V = IR \quad \text{potential difference (V) = current (A) } \times \text{ resistance } (\Omega)$$

You have already used this. On Friday you measured V and I for a length of nichrome wire and calculated $R = V/I$ — that is this equation, rearranged.

On terminology: the specification says *potential difference*, and exam questions will use it. *Voltage* means the same thing and you will not lose a mark for it, but get used to writing potential difference.

Worked example

A resistor has a resistance of $18\ \Omega$. The current through it is $0.25\ \text{A}$. Calculate the potential difference across it.

$$V = IR = 0.25 \times 18 = \mathbf{4.5\ V}$$

And rearranged: a pd of $6\ \text{V}$ across a $24\ \Omega$ resistor gives $I = V/R = 6/24 = 0.25\ \text{A}$.

7) **Calculate** the potential difference across a $12\ \Omega$ resistor carrying a current of $0.50\ \text{A}$.

$$V = IR = 0.50 \times 12 = \mathbf{6.0\ V}$$

8) **Calculate** the current through a $40\ \Omega$ resistor when the pd across it is $10\ \text{V}$.

$$I = V/R = 10/40 = \mathbf{0.25\ A}$$

9) **Calculate** the resistance of a component when a pd of $9.0\ \text{V}$ drives a current of $0.30\ \text{A}$ through it.

$$R = V/I = 9.0/0.30 = \mathbf{30}$$

10) **Two** components have the same potential difference across them. Component A has twice the resistance of component B. Compare the currents through them.

The current through A is half the current through B. For a fixed pd, current is inversely proportional to resistance — doubling the resistance halves the current.

Where this is going. You now have both equations the topic is built on: $Q = It$ and $V = IR$. Next lesson we ask a question you already have half an answer to — **does R stay the same when the current changes?** For the wire on Friday it did. For a lamp, it does not, and the graph is how you see it.

Before you go: the worksheet is the practice for $Q = It$ — eleven questions, and question 9 is the one that catches people. The past-paper sheet is your homework, due **Wednesday 23 September**.